Why did this happen? We ask ourselves this question a lot. About big things, such as the outcome of an election, or smaller things, such as a flat tire on our bike. What answer we give matters. It can mean the difference between winning and losing the next time, and avoiding another bike repair. Answering why-questions requires causal knowledge. A *why*-question wants a *because*-answer. **I study people's causal understanding of the world**. My work is guided by the idea that people build *mental models*. To be intelligent, to have good common-sense, is to be an expert model builder, and model user. My work shows that people's mental models mirror important aspects of the world's causal structure. This causal understanding allows them to predict what will happen next, infer what happened in the past, and simulate how things could have turned out differently – a capacity that is critical for answering questions about why something happened.

My work takes the foundational idea of a mental model and makes it precise and testable. I formalize mental models as computational models: computer programs that make quantitative predictions about different behavioral signatures, including judgments, decisions, language use, reaction times, and eye-movements. I design novel experimental paradigms to test the models' predictions against the behavior of human adults and children. Better understanding how people understand the world is important. Whether you're a lawyer, a teacher, a parent, or an AI system, you need to understand how your explanations affect others' causal understanding of the world. Analyzing "understanding" in computational terms is helpful because it allows knowledge to cross academic boundaries. Thanks to formal and computational work on causal inference, psychologists, philosophers, computer scientists, legal scholars, and many others now have a common language to communicate their causal models to each other. My work builds on this language, refines it, and applies it to distinctly psychological phenomena, focusing specifically on people's causal understanding of the physical and social domains. In the following, I share some of what I've discovered so far, organized around three core themes: 1) how people infer *what* happened, 2) understand *why* it happened, and 3) learn from *others' explanations* about the what and why.

Inferring what happened People have a remarkable ability to infer the hidden causes of things. From muddy footprints on the floor, we can figure out what happened and who did it. How do we achieve this impressive feat? I hypothesize that we do so by mentally simulating possible outcomes from an internal model of the world that represents how causes generate effects. Inverting this causal model allows us to reason from effect to cause, just like a detective who infers a culprit from a crime scene, or a doctor who infers a disease from a symptom.

Physical reasoning. Imagine a box with three holes at the top and some obstacles inside [30, 6, 46]. If you saw a ball being dropped in one of the holes, how would you predict where it ends up? We show that participants do so by mentally simulating different possible trajectories that the ball could take. We model people's causal knowledge of the domain, and their uncertainty about it, by using a physics engine and injecting noise into it. Sampling repeatedly from this noisy simulator yields a probability distribution of possible outcomes that closely matches participants' beliefs about where the ball will land. This paradigm allows us to study how people combine visual and auditory evidence to infer what happened. To do so, we cover up the box first so that participants only hear the sounds that the ball makes as it's dropped. Then, we reveal the cover, show where the ball ended up, and ask participants in which hole it was dropped. Some prior work has treated multi-modal inference as a merely statistical problem. In contrast, my work reveals that correctly integrating visual and auditory evidence to figure out what happened in this task requires causal knowledge. We developed a model that not only predicts participants' judgments but also captures the underlying cognitive process – mental simulation – as revealed through reaction times and eye-movements [6]. Participants focus on the most diagnostic features of the scene, such as places where the ball would collide with the obstacles. In other work, we used this model and paradigm to study the development of multi-modal inference [46]. Whereas 3- to 5-year-old children mostly guess that the ball was dropped from the hole closest to where it landed, 6 and 7-year-olds' choices are increasingly consistent with physical simulation, and 8-year-olds begin to integrate auditory information into their inferences, too.

Social reasoning. People use their mental models not only to simulate the physical world but also how people interact with it, and with one another [57, 52]. Imagine that you share an apartment with Alice and Bob. One day, your prized sandwich is missing from the fridge. Who took it? Based on the steps you heard that night, and the crumbs on the floor you see the next day, you can make an educated guess. We create scenarios like these in our experiments and find that participants' inferences are well-captured by a computational model that simulates how agents interact with their environment. This model uses the auditory and visual evidence that actions leave behind to figure out who the culprit was. While this kind of reasoning comes intuitively to people, current AI systems struggle with it [57, 33].

Understanding why it happened Knowing what happened is good – knowing *why* it happened is better. Answering why-questions about particular events often requires counterfactual thinking. If someone says "this happened because of that", a natural interpretation of what "because" means, is that "this" would not have happened without "that". Whereas prior work has questioned the role of counterfactuals in causal explanations [1, 18], my work demonstrates that counterfactual thoughts naturally come to people's minds when answering why-questions, both in the physical and social domains.

Physical reasoning. To better understand the cognitive mechanisms that support people's causal judgments, I have developed the counterfactual simulation model (CSM; [23, 20, 21, 22]). The CSM quantitatively captures people's causal judgments about dynamic physical events [23], causation by omission (where one event happened because another one didn't [31]), and physical support [62]. For example, when asked whether ball A caused ball B to go through a gate, people don't just look at what actually happened. They mentally simulate what would have happened if ball A hadn't been there [29]. While prior work failed to appreciate the distinction between hypotheticals ("Would B go through without A?") and counterfactuals ("Would B have gone through without A?"), I have demonstrated that counterfactual simulations are critical for explaining people's causal judgments, and that hypotheticals don't suffice [17]. The CSM not only captures how people judge causation but also how they talk about it [5]. The words we choose matter. "Killing someone" is not the same as "causing them to die". To explain how people use and interpret different causal expressions (including "caused", "enabled", "affected", and "made no difference"), the CSM simulates counterfactual contrasts to determine whether and how a candidate cause affected the outcome. Combinations of these counterfactual contrasts define the literal meaning of the causal expressions. For something to have "enabled" the outcome, it needs to have made a difference to *whether* it happened, whereas to have "caused" the outcome, it needs to have affected *how* it happened, too. Pragmatic principles then guide what expression a speaker uses in context so that the listener gets the most information about what happened. This model accurately captures what causal expressions people use, and what inferences adults [5] and children [49, 48] make based on what another person said.

Social reasoning. Judgments of causation are closely linked to attributions of responsibility. Part of what it means to be responsible for an outcome is to have played a causal role in bringing it about. This matters, for example, when people attribute responsibility to individuals in groups. Combining the contributions of many individuals often leads to overdetermined outcomes, such that the same thing would have happened even if any one person had acted differently. Prior work argued that such situations of overdetermination mean trouble for counterfactual theories. However, I have shown that across a wide range of experimental paradigms, a more sophisticated counterfactual model that considers how close a person's contribution was to having been pivotal for the outcome predicts people's responsibility judgments very well [25, 42, 63, 26, 27, 41, 2, 28]. Counterfactual theories require comparing what actually happened with an alternative [8]. With what though? The law uses different counterfactual tests [41, 36]: the "but for" test considers what would have happened if the person had not acted, whereas the "reasonable person" test considers what another person would have done. My work shows that people's responsibility judgments are sensitive to different counterfactual contrasts. Depending on the context, the same contribution is viewed as more or less responsible based on how difficult it would have been to replace [58]. Of course, there is more to what it means to be responsible than having caused the outcome. People also care about what the action reveals about the person [43, 16]. I extend the Bayesian theory of mind framework to capture this. My work has shown that people's responsibility judgments systematically depend on inferences about an agent's desires [51], intentions [37, 59], and capabilities [19, 32, 31].

Learning from others' explanations When people give explanations of what happened and why, they mention only a small subset of the many factors that contributed to the outcome. For example, they may attribute the election outcome to a particular demographic group that didn't vote as expected. Prior work argued that people generally prefer to cite abnormal events in explanations. In contrast, I have shown that people sometimes prefer normal events instead [39, 24]. What unifies these conflicting findings is that people prefer explanations that communicate good interventions [34]. Depending on the causal structure of the situation, it's better to target abnormal or normal events. Because of the systematic way in which people use their mental models to explain what happened, an explanation reveals much more than what it communicates literally. "B happened because of A" says more than "A caused B". Depending on the context, this explanation reveals the causal structure, the normality of the events, or the speaker's goals [35, 60]. In fact, even young children draw sophisticated inferences from others' explanations [3]. Relatedly, because people's intuitions of who should be held responsible are shared, knowing who was blamed (or praised) and how much, provides important clues about what happened and why [9].

Summary and future outlook The mental model is a powerful metaphor. My work shows that it has teeth. Assuming that people build mental models – ones that reflect the causal structure of the world and the way people interact with it – unifies a host of psychological phenomena: how people infer what happened from multi-modal evidence, how they judge causation and assign responsibility, and how they give and learn from explanations. Comparing the behavioral signatures of adult mental models with those of children [38, 7, 40, 3, 48, 46], and of current AI systems [14, 15, 11, 33, 45], yields insights into the cognitive mechanisms that power our intelligence [56, 55, 54, 12, 13, 4, 36, 10, 53].

One such mechanism – counterfactual thinking – may be a uniquely human capability. My research shows that in order to predict how people judge causation and assign responsibility, you have to assume that they think counterfactually. To demonstrate this convincingly, my work has focused on well-controlled settings. Going forward, I want to study these phenomena in more complex environments. I want to understand how people build abstract causal models that are tailored to their goals [50]. And I want to explore when and how counterfactual thinking impacts learning and decision-making [47, 61, 44]. Being able to play out different scenarios of what could have happened might help explain how people can learn so much from what they experience. Being able to anticipate one's own regret and the responsibility judgments of others, might lead to richer and more accurate models of decision-making in social contexts.

Teaching, mentoring, and building community My teaching, mentoring, and community-building efforts are guided by one core principle: *empowerment*. As a **teacher**, I empower my grad stats students by giving them tools to simulate and visualize data – tools that allow them to understand frequentist and Bayesian concepts, and to formalize their hypotheses as statistical models. In the psych honors class, I help create a community of scholars who support each other in their growth as scientific thinkers. As a **mentor**, I empower my lab members to produce excellent research, and to have fun doing so. Because we care that our work is reproducible and transparent we preregister it, use open-source software, and share our materials. As a **community builder**, I participate in community outreach programs to empower new minds to experience the joys of psychology. We share our lab's work with the public through press releases, blog posts, and social media. To build community within the department, I run a meditation group that meets every Wednesday morning to meditate and drink tea afterward. Namaste!

References

[1] Alicke, M. D., Mandel, D. R., Hilton, D., Gerstenberg, T., & Lagnado, D. A. (2015). Causal conceptions in social explanation and moral evaluation: A historical tour. *Perspectives on Psychological Science*, 10(6):790–812.

[2] Allen, K., Jara-Ettinger, J., Gerstenberg, T., Kleiman-Weiner, M., & Tenenbaum, J. B. (2015). Go fishing! responsibility judgments when cooperation breaks down. In D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, T. Matlock, C. D. Jennings, & P. P. Maglio, eds., *Proceedings of the 37th Annual Conference of the Cognitive Science Society*, pages 84–89. Cognitive Science Society, Austin, TX.

[3] Amemiya, J., Heyman, G. D., & Gerstenberg, T. (2024). Children use disagreement to infer what happened. *Cognition*.

[4] Andukuri, C., Fränken, J.-P., Gerstenberg, T., & Goodman, N. D. (2024). STaR-GATE: Teaching Language Models to Ask Clarifying Questions. *Conference on Language Modeling*. Http://arxiv.org/abs/2403.19154.

[5] Beller, A. & Gerstenberg, T. (2025). Causation, meaning, and communication. *Psychological Review*.

[6] Beller, A., Xu, Y., Linderman, S., & Gerstenberg, T. (2022). Looking into the past: Eye-tracking mental simulation in physical inference. *Cognitive Science Proceedings*.

[7] Bridgers, S., Yang, C., Gerstenberg, T., & Gweon, H. (2020). Whom will Granny thank? Thinking about what could have been informs children's inferences about relative helpfulness. In *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*.

[8] Brockbank, E., Yang, J., Govil, M., Fan, J. E., & Gerstenberg, T. (2024). Without his cookies, he's just a monster: A counterfactual simulation model of social explanation. In L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, & E. Hazeltine, eds., *Proceedings of the 46th Annual Conference of the Cognitive Science Society*.

[9] Davis, Z. J., Allen, K. R., Kleiman-Weiner, M., Jara-Ettinger, J., & Gerstenberg, T. (2025). Inference from social evaluation. *Journal of Personality and Social Psychology*.

[10] Du, M., Khazatsky, A., Gerstenberg, T., & Finn, C. (2024). To Err is Robotic: Rapid Value-Based Trial-and-Error during Deployment. *arXiv*. Http://arxiv.org/abs/2406.15917.

[11] Fränken, J.-P., Khawaja, A., Gandhi, K., Moore, J., Goodman, N. D., & Gerstenberg, T. (2023). Off the rails: Procedural dilemma generation for moral reasoning. In *AI Meets Moral Philosophy and Moral Psychology Workshop (NeurIPS 2023)*.

[12] Fränken, J.-P., Kwok, S., Ye, P., Gandhi, K., Arumugam, D., Moore, J., Tamkin, A., Gerstenberg, T., & Goodman, N. D. (2023). Social contract ai: Aligning ai assistants with implicit group norms. In *Socially Responsible Language Modelling Research Workshop (NeurIPS 2023)*.

[13] Fränken, J.-P., Zelikman, E., Rafailov, R., Gandhi, K., Gerstenberg, T., & Goodman, N. D. (2024). Self-supervised alignment with mutual information: Learning to follow principles without preference labels. In *Advances in Neural Information Processing Systems*. Http://arxiv.org/abs/2404.14313.

[14] Gandhi*, K., Fränken*, J.-P., Gerstenberg, T., & Goodman, N. D. (2023). Understanding social reasoning in language models with language models. In *Advances in Neural Information Processing Systems*.

[15] Gandhi, K., Lynch, Z., Fränken, J.-P., Patterson, K., Wambu, S., Gerstenberg, T., Ong, D. C., & Goodman, N. D. (2024). Human-like affective cognition in foundation models. *arXiv*. Https://arxiv.org/abs/2409.11733.

[16] Gershman*, S. J., Gerstenberg*, T., Baker, C. L., & Cushman, F. (2016). Plans, habits, and theory of mind. *PLoS ONE*, 11(9):e0162246.

[17] Gerstenberg, T. (2022). What would have happened? counterfactuals, hypotheticals, and causal judgments. *Philosophical Transactions of the Royal Society B: Biological Sciences*.

[18] Gerstenberg, T. (2024). Counterfactual simulation in causal cognition. *Trends in Cognitive Sciences*.

[19] Gerstenberg, T., Ejova, A., & Lagnado, D. A. (2011). Blame the skilled. In C. Carlson, C. Hölscher, & T. Shipley, eds., *Proceedings of the 33rd Annual Conference of the Cognitive Science Society*, pages 720–725. Austin, TX: Cognitive Science Society.

[20] Gerstenberg, T., Goodman, N. D., Lagnado, D. A., & Tenenbaum, J. B. (2012). Noisy Newtons: Unifying process and dependency accounts of causal attribution. In N. Miyake, D. Peebles, & R. P. Cooper, eds., *Proceedings of the 34th Annual Conference of the Cognitive Science Society*, pages 378–383. Austin, TX: Cognitive Science Society.

[21] Gerstenberg, T., Goodman, N. D., Lagnado, D. A., & Tenenbaum, J. B. (2014). From counterfactual simulation to causal judgment. In P. Bello, M. Guarini, M. McShane, & B. Scassellati, eds., *Proceedings of the 36th Annual Conference of the Cognitive Science Society*, pages 523–528. Cognitive Science Society, Austin, TX.

[22] Gerstenberg, T., Goodman, N. D., Lagnado, D. A., & Tenenbaum, J. B. (2015). How, whether, why: Causal judgments as counterfactual contrasts. In D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, J. Matlock, T., C. D., & P. P. Maglio, eds., *Proceedings of the 37th Annual Conference of the Cognitive Science Society*, pages 782–787. Cognitive Science Society.

[23] Gerstenberg, T., Goodman, N. D., Lagnado, D. A., & Tenenbaum, J. B. (2021). A counterfactual simulation model of causal judgments for physical events. *Psychological Review*, 128(6):936–975.

[24] Gerstenberg, T. & Icard, T. F. (2020). Expectations affect physical causation judgments. *Journal of Experimental Psychology: General*, 149(3):599–607.

[25] Gerstenberg, T. & Lagnado, D. A. (2010). Spreading the blame: The allocation of responsibility amongst multiple agents. *Cognition*, 115(1):166–171.

[26] Gerstenberg, T. & Lagnado, D. A. (2012). When contributions make a difference: Explaining order effects in responsibility attributions. *Psychonomic Bulletin & Review*, 19(4):729–736.

[27] Gerstenberg, T. & Lagnado, D. A. (2014). Attributing responsibility: Actual and counterfactual worlds. In J. Knobe, T. Lombrozo, & S. Nichols, eds., *Oxford Studies in Experimental Philosophy*, volume 1, pages 91–130. Oxford University Press.

[28] Gerstenberg, T., Lagnado, D. A., & Zultan, R. (2023). Making a positive difference: Criticality in groups. *Cognition*.

[29] Gerstenberg, T., Peterson, M. F., Goodman, N. D., Lagnado, D. A., & Tenenbaum, J. B. (2017). Eye-tracking causality. *Psychological Science*, 28(12):1731–1744.

[30] Gerstenberg, T., Siegel, M. H., & Tenenbaum, J. B. (2021). What happened? Reconstructing the past from vision and sound. *PsyArXiv*.

[31] Gerstenberg, T. & Stephan, S. (2021). A counterfactual simulation model of causation by omission. *Cognition*, 216:104842.

[32] Gerstenberg, T., Ullman, T. D., Nagel, J., Kleiman-Weiner, M., Lagnado, D. A., & Tenenbaum, J. B. (2018). Lucky or clever? From expectations to responsibility judgments. *Cognition*, 177:122–141.

[33] Jin*, E., Huang*, Z., Fränken, J.-P., Liu, W., Cha, H., Brockbank, E., Wu, S., Zhang, R., Wu, J., & Gerstenberg, T. (2024). MARPLE: A Benchmark for Long-Horizon Inference. In *Advances in Neural Information Processing Systems*. Http://arxiv.org/abs/2410.01926.

[34] Kirfel, L., Harding, J., Shin, J., Xin, C., Icard, T., & Gerstenberg, T. (2024). Do as I explain: Explanations communicate optimal interventions. In L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, & E. Hazeltine, eds., *Proceedings of the 46th Annual Conference of the Cognitive Science Society*.

[35] Kirfel, L., Icard, T. F., & Gerstenberg, T. (2022). Inference from explanation. *Journal of Experimental Psychology: General*.

[36] Kirfel, L., MacCoun, R. J., Icard, T., & Gerstenberg, T. (2025). When AI meets counterfactuals: The ethical implications of counterfactual world simulation models. *AI and Ethics*.

[37] Kleiman-Weiner, M., Gerstenberg, T., Levine, S., & Tenenbaum, J. B. (2015). Inference of intention and permissibility in moral decision making. In D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, J. Matlock, T., C. D., & P. P. Maglio, eds., *Proceedings of the 37th Annual Conference of the Cognitive Science Society*, pages 1123–1128. Cognitive Science Society, Austin, TX.

[38] Kominsky, J. F., Gerstenberg, T., Pelz, M., Singmann, H., Sheskin, M., & Keil, F. (2019). The trajectory of counterfactual simulation in development. In A. Goel, C. Seifert, & C. Freksa, eds., *Proceedings of the 41st Annual Conference of the Cognitive Science Society*, pages 2044–2050. Cognitive Science Society, Montreal, QB.

[39] Kominsky, J. F., Phillips, J., Gerstenberg, T., Lagnado, D. A., & Knobe, J. (2015). Causal superseding. *Cognition*, 137:196–209.

[40] Koskuba, K., Gerstenberg, T., Gordon, H., Lagnado, D. A., & Schlottmann, A. (2018). What's fair? how children assign reward to members of teams with differing causal structures. *Cognition*, 177:234–248.

[41] Lagnado, D. A. & Gerstenberg, T. (2017). Causation in legal and moral reasoning. In M. Waldmann, ed., *Oxford Handbook of Causal Reasoning*, pages 565–602. Oxford University Press.

[42] Lagnado, D. A., Gerstenberg, T., & Zultan, R. (2013). Causal responsibility and counterfactuals. *Cognitive Science*, 47:1036–1073.

[43] Langenhoff, A. F., Wiegmann, A., Halpern, J. Y., Tenenbaum, J. B., & Gerstenberg, T. (2021). Predicting responsibility judgments from dispositional inferences and causal attributions. *Cognitive Psychology*, 129:101412.

[44] Nam, A., Hughes, C., Icard, T., & Gerstenberg, T. (2023). Show and tell: Learning causal structures from observations and explanations. In M. B. Goldwater, F. Anggoro, B. Hayes, & D. C. Ong, eds., *Proceedings of the 45th Annual Conference of the Cognitive Science Society*.

[45] Nie, A., Zhang, Y., Amdekar, A., Piech, C. J., Hashimoto, T., & Gerstenberg, T. (2023). MoCa: Measuring human-language model alignment on causal and moral judgment tasks. In *Advances in Neural Information Processing Systems*.

[46] Outa*, J., Zhou*, X. J., Gweon, H., & Gerstenberg, T. (2022). Stop, children what's that sound? multi-modal inference through mental simulation. *Cognitive Science Proceedings*.

[47] Petrova, K., Gross, J., & Gerstenberg, T. (2025). Learning from 'what might have been': A Bayesian model of learning from regret. In D. Barner, C. Walker, A. Ruggeri, & N. Bramley, eds., *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

[48] Rose, D., Hou, C., Nichols, S., Gerstenberg, T., & Markman, E. M. (2025). Cause and fault in development. In D. Barner, C. Walker, A. Ruggeri, & N. Bramley, eds., *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

[49] Rose, D., Zhang, S., Nichols, S., Markman, E., & Gerstenberg, T. (2025). Cause and burn in development. *PsyArXiv*. Https://osf.io/preprints/psyarxiv/aw834.

[50] Shin, S. M. & Gerstenberg, T. (2023). Learning what matters: Causal abstraction in human inference. In M. B. Goldwater, F. Anggoro, B. Hayes, & D. C. Ong, eds., *Proceedings of the 45th Annual Conference of the Cognitive Science Society*.

[51] Sosa, F. A., Ullman, T., Tenenbaum, J. B., Gershman, S. J., & Gerstenberg, T. (2021). Moral dynamics: Grounding moral judgment in intuitive physics and intuitive psychology. *Cognition*, 217:104890.

[52] Teo, V., Wu, S. A., Brockbank, E., & Gerstenberg, T. (2025). Leave a trace: Recursive reasoning about deceptive behavior. In D. Barner, C. Walker, A. Ruggeri, & N. Bramley, eds., *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

[53] Tsirtsis, S., Gomez-Rodriguez, M., & Gerstenberg, T. (2024). Towards a computational model of responsibility judgments in sequential human-AI collaboration. In L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, & E. Hazeltine, eds., *Proceedings of the 46th Annual Conference of the Cognitive Science Society*.

[54] Vasconcelos, H., Jörke, M., Grunde-McLaughlin, M., Gerstenberg, T., Bernstein, M. S., & Krishna, R. (2023). Explanations can reduce overreliance on ai systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW1):1–38.

[55] Vodrahalli, K., Daneshjou, R., Gerstenberg, T., & Zou, J. (2022). Do humans trust advice more if it comes from AI? an analysis of human-AI interactions. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pages 763–777.

[56] Vodrahalli, K., Gerstenberg, T., & Zou, J. Y. (2022). Uncalibrated models can improve human-AI collaboration. *Advances in Neural Information Processing Systems*, 35:4004–4016.

[57] Wu, S. A., Brockbank, E., Cha, H., Fränken, J.-P., Jin, E., Huang, Z., Liu, W., Zhang, R., Wu, J., & Gerstenberg, T. (2024). Whodunnit? Inferring what happened from multimodal evidence. In L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, & E. Hazeltine, eds., *Proceedings of the 46th Annual Conference of the Cognitive Science Society*.

[58] Wu, S. A. & Gerstenberg, T. (2023). If not me, then who? Responsibility and replacement. *Cognition*.

[59] Wu, S. A., Sridhar, S., & Gerstenberg, T. (2023). A computational model of responsibility judgments from counterfactual simulations and intention inferences. In M. B. Goldwater, F. Anggoro, B. Hayes, & D. C. Ong, eds., *Proceedings of the 45th Annual Conference of the Cognitive Science Society*.

[60] Xiang, Y., Gershman, S. J., & Gerstenberg, T. (2024). A signaling theory of self-handicapping. *PsyArXiv*. Https://osf.io/preprints/psyarxiv/84tvm.

[61] Yang, J. & Gerstenberg, T. (2025). Learning to plan from observed and counterfactual experiences. In D. Barner, C. Walker, A. Ruggeri, & N. Bramley, eds., *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

[62] Zhou, L., Smith, K. A., Tenenbaum, J. B., & Gerstenberg, T. (2023). Mental Jenga: A counterfactual simulation model of causal judgments about physical support. *Journal of Experimental Psychology: General*.

[63] Zultan, R., Gerstenberg, T., & Lagnado, D. A. (2012). Finding fault: Counterfactuals and causality in group attributions. *Cognition*, 125(3):429–440.